MEMORY SYSTEMS & CACHE QUIZ

Advanced Computer Architecture

Name: ______________________________ **ERP ID:** ________________

Instructions: Solve the following 20 questions. Show calculations where necessary.

1. In a cache system, if the hit time is 2 cycles, the miss rate is 10%, and the miss penalty is 50 cycles, what is the Average Memory Access Time (AMAT)?

A. 5 cycles

B. 7 cycles

C. 52 cycles

D. 12 cycles

2. A cache has 1024 sets and is 4-way set-associative. If the block size is 16 bytes, what is the total cache capacity?

A. 128 KB

B. 32 KB

C. 64 KB

D. 16 KB

3. A 4-way set-associative cache has a total capacity of 16 KB and a block size of 64 bytes. How many sets are in this cache?

A. 32 sets

B. 128 sets

C. 256 sets

D. 64 sets

4. In a 32-bit address, if the index uses bits [13:6] and the offset uses bits [5:0], how many total blocks are in the cache if it is 2-way set-associative?

A. 1024

B. 512

C. 128

D. 256

5. Which address bits are used for the 'Offset' in a system with 128-byte cache blocks?

A. Bits [7:0]

B. Bits [6:0]

C. Bits [4:0]

D. Bits [5:0]

6. A system uses a 32-bit address and a 16 KB direct-mapped cache with 32-byte blocks. How many bits are required for the Tag field?

A. 12 bits

B. 15 bits

C. 18 bits

D. 21 bits

7. If the associativity of a cache is doubled while keeping the total size and block size the same, what happens to the number of index bits?

A. It decreases by 2 bits.

B. It decreases by 1 bit.

C. It remains unchanged.

D. It increases by 1 bit.

8. If a 2-way set-associative cache is changed to a direct-mapped cache of the same total size and block size, how does the index field change?

A. It decreases by 1 bit.

B. It increases by 1 bit.

C. It stays the same.

D. It increases by 2 bits.

9. Which of the following is a disadvantage of increasing cache block size?

A. It decreases spatial locality.

B. It increases the compulsory miss rate.

C. It requires more bits for the tag field.

D. It increases the miss penalty due to longer transfer times.

10. If a program performs 2000 accesses with 100 misses, and the hit time is 1ns with a 40ns miss penalty, what is the total execution time for these accesses?

A. 82000 ns

B. 4000 ns

C. 6000 ns

D. 2000 ns

11. A cache has 512 sets and is 2-way set-associative. If the block size is 32 bytes, what is the total cache capacity?

A. 16 KB

B. 64 KB

C. 32 KB

D. 8 KB

12. Which type of locality is being exploited when a loop accesses the same variable multiple times in a short interval?

A. Temporal Locality

B. Conflict Locality

C. Spatial Locality

D. Sequential Locality

13. In a 32-bit RISC-V address, if the index uses bits [11:4] and the offset uses bits [3:0], how many sets are in the cache?

A. 16

B. 128

C. 512

D. 256

14. What is the primary function of a Write Buffer in a 'Write-Through' cache?

A. To track which blocks are the least recently used.

B. To store blocks evicted from a victim cache.

C. To store data during a read miss.

D. To allow the CPU to continue without waiting for main memory to finish a write.

15. Which replacement policy is most complex to implement in hardware as associativity increases?

A. LRU (Least Recently Used)

B. NMRU (Not Most Recently Used)

C. Random

D. FIFO (First-In, First-Out)

16. If a cache has 1024 blocks and is direct-mapped, how many index bits are required?

A. 1024 bits

B. 10 bits

C. 8 bits

D. 12 bits

17. Under which condition does a 'Conflict Miss' occur in a fully associative cache?

A. When the tag matches but the valid bit is zero.

B. When the block is accessed for the first time.

C. When two addresses map to the same set.

D. Conflict misses do not occur in fully associative caches.

18. A CPU has an L1 hit time of 1ns and an L1 miss rate of 10%. The L2 hit time is 10ns and the L2 miss rate is 50%. The main memory access time is 100ns. What is the total AMAT?

A. 111 ns

B. 11 ns

C. 7 ns

D. 6.1 ns

19. How many index bits are used in a 128 KB 8-way set-associative cache with 64-byte blocks?

A. 8 bits

B. 14 bits

C. 10 bits

D. 12 bits

20. For a 64 KB cache with 4-way associativity and 64-byte blocks, what is the size of the 'Tag' field in a 32-bit address?

A. 16 bits

B. 20 bits

C. 18 bits

D. 14 bits

End of Examination